Phase 2

Requirements engineering

1. Stakeholder analysis and conflict resolution

1.1 Stakeholders

Stakeholder	Main interest
Principal	Wants a simple system and approved marks before students see them.
Lecture	Wants to create assignment, give extensions, mark work and give feedback.
Head of Department	Wants to check marks and apply the late submission rules.
Student	Wants to submit and resubmit work, and see marks and feedback.
Administrator	Wants to manage users, courses and enrolments.
IT Officer	Wants a secure system that works offline and runs well on old computers

1.2 Conflicts and how they are resolved.

Conflict or problem	Resolution
Lecturers want marks shown immediately, but the Principal wants them checked first.	Students can only see marks after the HoD approves and releases them.
The HoD wants late work penalized, while students do not want it rejected.	Late work is accepted, but the agreed penalty is applied.
Students want to resubmit, but lecturers need to mark the correct version.	Students can resubmit before the deadline, and the latest valid submission is marked.
Students may need extra time.	Lecturers can grant individual extensions and record the new deadline.
The school needs security, but its computers are old and the internet is unreliable.	The system will work offline, protect passwords and be tested for performance.

Assumptions: work submitted within an approved extension is treated as on time. Late work loses 10% of the maximum mark per day, for to three days. After three days, it receives zero. The HoD must approve marks before release.

2. Software Requirements specification

Priority: **M= Must have**, S= **Should have**, C= **Could have**.

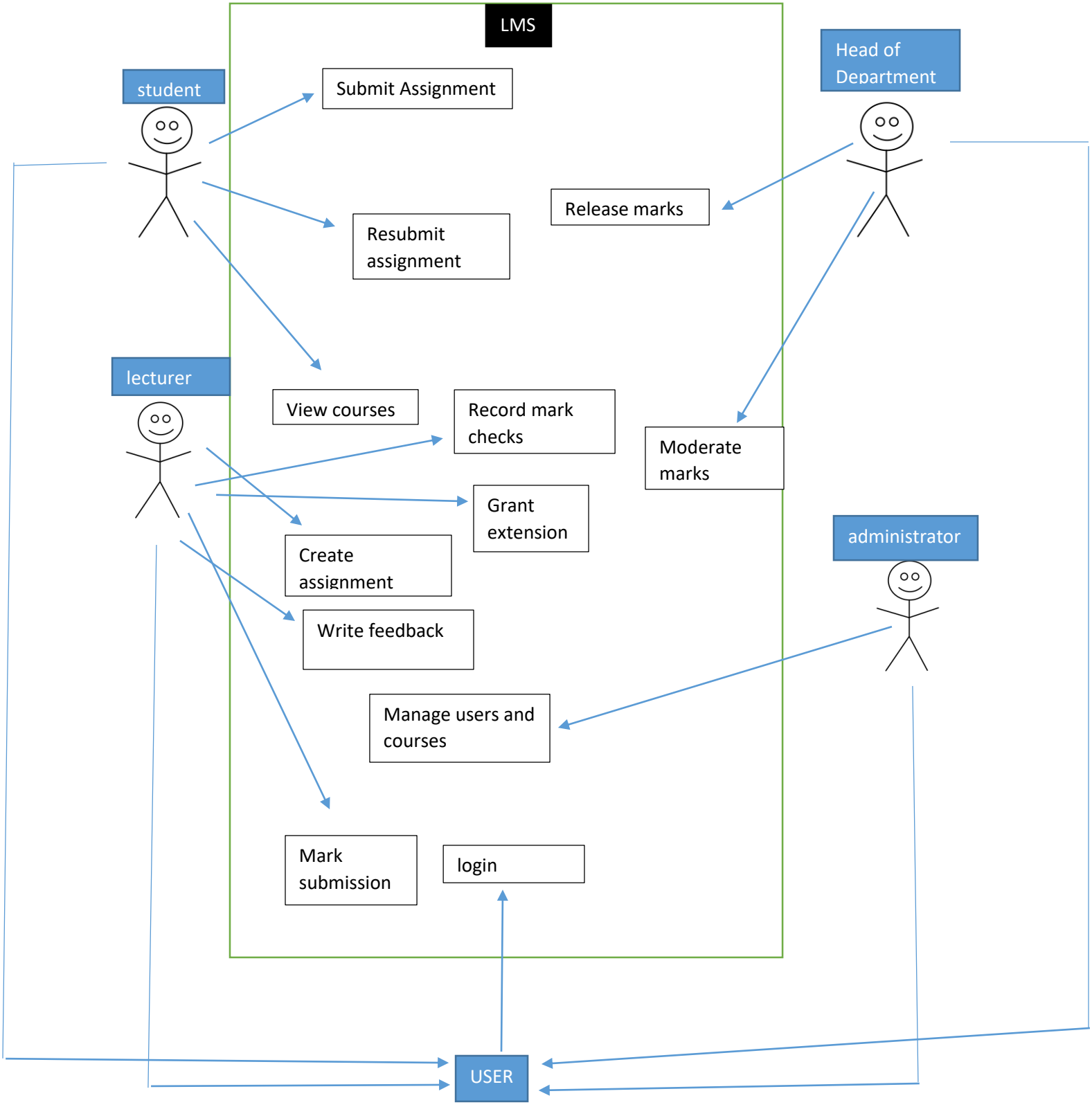
ID	Requirement	Priority
FR-01	The system should allow users to login with a username and password	M
FR-02	The system shall allow the administrator to create, edit and deactivate user accounts	M
FR-03	The system shall allow the administrator to manage department and courses	M
FR-04	The system shall allow the administrator to assign lecturers to courses.	M
FR-05	The system shall allow the administrator to enroll students.	M
FR-06	The system shall allow lecturers to create assignments with and maximum marks.	M
FR-07	The system shall allow lecturers to grant and record individual extensions.	M
FR-08	The system shall allow students to submit and resubmit work before the applicable deadline.	M
FR-09	The system shall record submissions as on time or late.	M
FR-10	The system shall allow students to view their enrolled courses and assignments.	M
FR-11	The system shall allow lecturers to mark submissions and provide written feedback.	M
FR-12	The system shall apply the agreed late-submission penalty automatically.	M

FR-13	The system shall allow HoDs to review and moderate marks in their departments.	M
FR-14	The system shall allow only the HoDs to approve and release marks to students.	M
FR-15	The system shall record every mark change, including who changed it, when and why	M
FR-16	The system shall allow students to view released marks before feedback.	M

Non-Functional requirements

ID	Requirements	priority
NFR-01	The system shall complete login within 2 seconds with 800 student records stored.	M
NFR-02	The system shall operate without an internet connection.	M
NFR-03	The system shall store passwords as secure hashes and not plain text.	M
NFR-04	The system shall prevent users from accessing functions outside their assigned roles in all access-control tests.	M
NFR05	The system shall complete course and assignment searches within 2 seconds with 800 student records stored.	S
NFR-06	The system shall pass all agreed critical tests before release.	M

3. Use case model


LMS
student
Submit Assignment
Resubmit assignment
Head of Department
Release marks
lecturer
View courses
Record mark checks
Moderate marks
Grant extension
Create assignment
Write feedback
administrator
Manage users and courses
Mark submission
login
USER

Fully dressed use case 1: submit assignment

Item	Description
User case ID	UC01
Actor	Student
Goal	Submit work for an assignment
Preconditions	The student is logged in and enrolled in the course
Trigger	The student selects an assignment
Main flow	1. The student opens the assignment. 2. The system displays its instructions and deadline. 3. The student selects and submits a file. 4. The system checks the deadline. 5. The system saves the submission and confirms success.
Alternative flow	The student resubmits before the dead line. The system saves the new version as the latest submission.
Exception flow	If the file cannot be saved, the system displays an error. If the deadline has passed, the system accepts the work as late only within the allowed period or an approved extension.
Post-condition	The submission is stored with its submission time and status.
Business rules	Extensions change the applicable deadline. Let penalties are applied according to the agreed rules.

Fully dressed use case 2: Mark submission

Item	Description
Use case ID	UC02
Actor	Lecturer
Goal	Give a student a mark and written feedback
Pre-conditions	The lecturer is logged in, assigned to the course and the student has submitted work.
Trigger	The lecturer opens a submission for marking
Main flow	1. The lecturer views the submission. 2. The lecturer enters a mark and feedback 3. The system checks that the mark is within the allowed range. 4. The system applies any late penalty. 5. The system saves the mark and feedback
Alternative flow	The lecturer corrects a mark and gives a reason for the change.
Exception flow	The system rejects marks outside the allowed range or refuses unauthorized access.
Post-conditions	The mark and feedback are saved for HoD review
Business rules	Students cannot see the mark until the HoD approves and releases it. Every mark change must be recorded with the user, time and reason.

Fully dressed use case 3: moderate and release marks

Item	Description
Use case ID	UC03
Actor	Head of department
Goal	Check marks and release approved results
Pre-conditions	The HoD is logged in and has access to the relevant department. Lecturers have entered marks.
Trigger	The HoD opens the marks awaiting moderation.
Main flow	1. The HoD views marks for the department. 2. The HoD approves the marks or returns them for correction. 3. The system records the decision. 4. The HoD releases approved marks 5. The system makes released results available to students.
Alternative flow	The HoD returns an incorrect mark to the lecturer with a reason. The lecturer corrects it and resubmits its moderation.
Exception flow	If the HoD is not authorized, the system denies access. If marks are not approved, the system prevents their release.

Post-conditions	Approved marks are released, or the marks needing correction remain released.
Business rules	Only an authorized HoD may moderate and release marks for their department, all mark changes and moderation decisions must be recorded.